BT3041:Course Project

Prediction of Systemic Lupus Erythematosus Disease Activity from Genomic Data

Aldis Daniel (ME22B070) R Karthik (BS22B022) Vasudha Kannan (BS22B040)
Hitesh Kandarpa (BS22B021)

ABSTRACT

Systemic Lupus Erythematosus (SLE) is a complex autoimmune disease with a strong genetic component and highly variable clinical features. Because of this heterogeneity, predicting disease status remains challenging using any single marker alone. In this project, we explore two complementary data sources , (1) clinical mutation data and (2) bulk gene-expression profiles to investigate whether they can be used to distinguish SLE cases from healthy controls. The general goal is to compare the predictive value of these two modalities and assess their potential to support more accurate disease classification.

INTRODUCTION

Systemic Lupus Erythematosus (SLE) is a systemic autoimmune disease characterized by loss of tolerance to self-antigens, auto-antibody production, and immune-mediated tissue damage that can affect multiple organs, including the skin, joints, kidneys, and central nervous system (Tufan, Petri, Chama, et al.,).

The accurate classification of SLE is clinically important because it influences the intensity of treatment, organ-specific monitoring, and prognosis, yet current criteria are still largely based on clinical and serological features rather than molecular data (Aringer, Costenbader, Daikh, et al.,). This motivates the development of data-driven machine-learning approaches that can leverage genetic and gene-expression information to refine disease-state prediction and improve patient stratification (Braun & et al.,).

In this work, we adopt a dual approach to predict disease state using both clinical mutation data and bulk gene-expression data. By analyzing these two modalities together, we aim to capture both inherited risk signals and downstream transcriptional changes associated with SLE.

DATASETS

Clinical Mutational Data

The SNP genotype data set was taken from (Rupasree et al.,) , a study designed to assess the contribution of genetic variation at the pathway-level to the susceptibility

to SLE in a South Indian cohort. It comprises 188 SLE-diagnosed individuals (178 females, 10 males) and 237 healthy controls (230 females, 7 males), totalling 425 subjects. Genotypes at 30 SNP loci were measured using PCR-based methods and are encoded as 0, 1, or 2, representing the number of copies of the alternate allele present (homozygous reference, heterozygous, and homozygous alternate respectively).

The 30 SNPs span four pathways that have been shown independently to affect in autoimmune pathogenesis. In addition to this, the data also includes age at sample collection and biological sex.

Gene Expression Data

The gene expression dataset was assembled by manually curating bulk profiles from the NCBI Gene Expression Omnibus (GEO). The primary study contributing samples is GSE10325 and includes measurements across $CD4^+$ T cells, $CD19^+$ B cells, and myeloid cells from both SLE patients and healthy donors. Additional studies were also incorporated, the full dataset is available in the github page for the project

The sample labels (Disease or Healthy) were derived from GEO metadata annotations embedded in the `title` and `characteristics_ch1` fields of each GSM entry. The complete data set before preprocessing contains raw probe-level expression values, metadata columns (GSE ID, GSM ID, GPL platform, title), and gene-level expression columns, with genes identified by their HGNC symbols. Invalid expression values, encoded as -1 in the raw files, indicate probe failures or curation artifacts and are treated as missing data throughout the pipeline.

METHODOLOGY

Gene Expression Data Profiling

Before any normalisation or analysis, we first characterised the structure of the raw expression data to understand what we were working with. The dataset was classified as bulk gene expression rather than single-cell. This distinction matters because bulk profiles reflect the aggregate transcriptional state of a mixed cell population, and any observed differences between disease and healthy samples represent a combination of true gene regulatory changes and shifts in cell-type composition.

Metadata columns were separated from gene expression columns at the loading stage. Columns corresponding to GSE ID, GSM ID, GPL platform, and title were retained as metadata but excluded from any numerical computation. The remaining columns, which correspond to individual gene probes, were examined for coverage: specifically, how many samples had a valid measurement for each gene, and what fraction of entries were flagged as missing (-1 or NaN). Genes where every sample was missing were dropped entirely at this stage.

We also examined the raw distribution of expression values before any transformation. Microarray probe intensities tend to span several orders of magnitude and are highly right-skewed, with a small number of highly expressed probes dominating the distribution. This property motivates the log transformation described in the next section. The per-sample expression profiles were also inspected for outlier samples, which were identified as those whose overall expression distribution deviated markedly from the rest of the cohort.

Data Normalisation

Given the multi-study, multi-platform origin of the expression data, The normalization was applied carefully to ensure that biological signal was preserved while technical variability was minimized. The full normalisation pipeline is as follows :

Step 1 : Invalid value handling. All entries equal to -1 were replaced with `NaN`. Genes for which every sample was missing were removed at this point. This is distinct from removing genes with only a few missing values, which were handled by imputation later.

Step 2 : Log$_2$ transformation. A $\log_2(x+1)$ transformation was applied to all remaining expression values.The pseudocount of 1 prevents the undefined values from settling at zero. This transformation compresses the dynamic range of the data, stabilises variance across the expression range, and brings the distribution closer to normality, which is a prerequisite for several downstream statistical tests.

Step 3 : Imputation of missing values. After log transformation, residual missing values were filled using the gene-wise median across all samples. Gene-wise median imputation was preferred over global or sample-wise strategies because it preserves the relative expression level of each gene and is robust to the asymmetric distributions that persist even after log transformation in some probes.

Step 4 : Z-score normalisation. Finally, each gene was standardised to zero mean and unit variance by subtracting the gene's mean expression and dividing by its standard deviation, computed across all samples. This per-gene normalisation removes systematic differences in probe sensitivity and hybridisation efficiency between genes, rendering expression values directly comparable in the same scale. After this step, each value represents the number of standard deviations by which a given sample's expression of a given gene deviates from the cohort average.

The output of this pipeline is a clean, fully normalised matrix suitable for differential expression testing and machine learning. The final dimensions of this matrix, along with QC plots, are reported in the Results section.

Analysis Methods

Genotype QC

Missingness was calculated for each of the 30 SNPs. Hardy-Weinberg equilibrium (HWE) was then checked in the control group , giving a p-value per variant. Any variant with HWE p < 0.001 was flagged as a possible genotyping error, but kept in the analysis since the deviation could also reflect a real biological signal

Per-SNP association

For each SNP, a χ^2 test of independence was applied to the 3×2 genotype-by-outcome contingency table. Nominal p-values were corrected for multiple testing across 30 variants using the Benjamini Hochberg (BH) procedure at FDR < 0.05.

Genetic encoding

Three inheritance models were evaluated for every SNP: *additive* (dose 0/1/2), *dominant* (any alternate allele $\rightarrow 1$), and *recessive* (homozygous alternate only $\rightarrow 1$). Each encoding was tested in logistic regression alongside age and sex as covariates.

Classification

L2-penalised logistic regression (LR; $C = 1$) and XGBoost (300 estimators, max depth 3, learning rate 0.03) were evaluated under stratified 5-fold cross-validation. Performance was assessed by ROC-AUC, F1, and accuracy on out-of-fold predictions. For the gene-expression modality, L1-penalised LR was used to select informative genes from the 13-feature matrix.

Interpretability

SHAP (SHapley Additive exPlanations) values were computed for the full-data XGBoost model to rank SNP contributions by mean |SHAP|.

Differential expression

Mann–Whitney U tests were applied per gene between SLE and Healthy Control samples. Median $\log_2$ fold-change (SLE $-$ HC) was used as an effect-size estimate. BH-FDR correction was applied across the 13 genes.

RESULTS

SNP QC and Variant Association

Overall missingness was low (mean $< 1\%$). HWE testing in controls flagged two variants with extreme deviations: MTRR A66G ($p = 1.84 \times 10^{-12}$) and TLR5 rs2072493 ($p = 1.52 \times 10^{-10}$). These deviations may reflect local genotyping difficulty effects rather than

true biological enrichment (Fig. 1, panel A).

Per-SNP χ^2 association testing identified four variants passing FDR < 0.05 (Fig. 1, panel B): MTRR A66G ($q = 0.0002$), CYP1A1M2 ($q = 0.014$), COMT ($q = 0.032$), and CYP1A1M4 ($q = 0.044$). Seven additional SNPs reached nominal significance ($p < 0.05$), including MTHFR ($p = 0.034$), OPN ($p = 0.032$), and TNFα ($p = 0.039$). The dominant folate-cycle signal in MTRR A66G is consistent with H2.

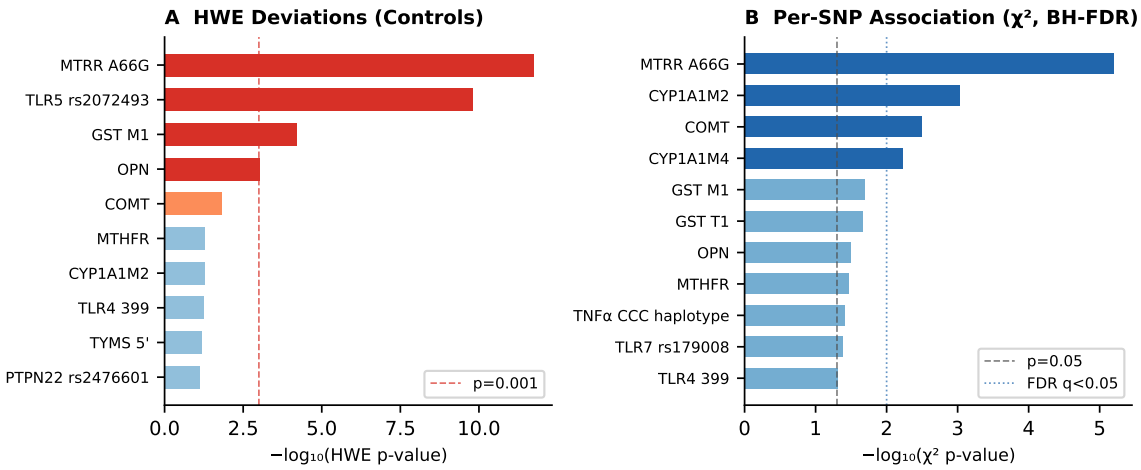


Figure 1: SNP quality control and association. (A) HWE p-values in controls ($-\log_{10}$ scale); dashed line at $p = 0.001$. (B) Per-SNP χ^2 association strength after BH-FDR correction; dark blue bars indicate $q < 0.05$.

Classification Performance

Table 1 summarises 5-fold cross-validated performance across all models. Age and sex alone yield AUC $= 0.634$, confirming that these covariates carry limited discriminative power. Adding SNP features under an additive encoding raises AUC to 0.662; switching to a dominant encoding further improves performance to 0.691 (LR), supporting H1. XGBoost under additive encoding achieves AUC $= 0.667$, which is slightly below the dominant LR and consistent with H3. The gene-expression L1-LR substantially outperforms all SNP models with AUC $= 0.807 \pm 0.025$ (H4 strongly supported). ROC curves are shown in Fig. 2.

Table 1: 5-fold CV performance summary. AUC reported as mean $\pm$ SD across folds.

Modality	Model	ROC-AUC	F1	Accuracy
SNP	Age + Sex (baseline)	0.634 ± 0.096	0.461	0.593
SNP	Additive + Age+Sex (LR)	0.662 ± 0.046	0.550	0.614
SNP	**Dominant + Age+Sex (LR)**	$\mathbf{0.691 \pm 0.059}$	**0.588**	**0.654**
SNP	Additive + Age+Sex (XGBoost)	0.667 ± 0.051	0.561	0.640
GEX	**13-gene L1 Logistic Regression**	$\mathbf{0.807 \pm 0.025}$	**0.827**	**0.740**

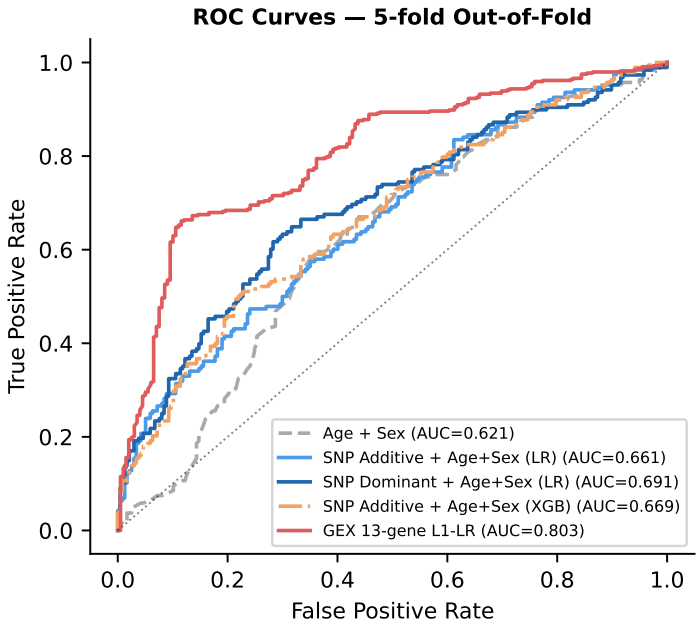


Figure 2: ROC curves (5-fold out-of-fold). All SNP model variants shown alongside the GEX baseline. The gene-expression classifier (red) substantially outperforms the best SNP model (dark blue).

Feature Importance and Differential Expression

SHAP analysis identifies age, OPN, MTRR A66G, COMT, and GST T1 as the top-5 SNP model contributors (Fig. 3, panel A), with age standing out as a strong non-genetic confound.

Of the 13 pathway genes, 9 are differentially expressed at FDR < 0.05 (Fig. 3, panel B). TLR2 is the most strongly upregulated gene ($\log_2$FC $= +2.39$, $q = 7 \times 10^{-22}$). MTR and MTRR are downregulated ($\log_2$FC $= -0.38$ and -0.42, $q < 10^{-16}$), reinforcing the folate-cycle signal seen in the SNP data. STAT4 is also upregulated ($\log_2$FC $= +0.96$, $q = 5 \times 10^{-13}$).

Hypotheses

- **H1:** Dominant encoding outperforms additive encoding for SNP-based classification.
- **H2:** Folate-pathway SNPs show the strongest association with SLE.
- **H3:** XGBoost will not substantially outperform logistic regression given the small sample size.
- **H4:** Gene expression data will be more discriminative than SNP data.

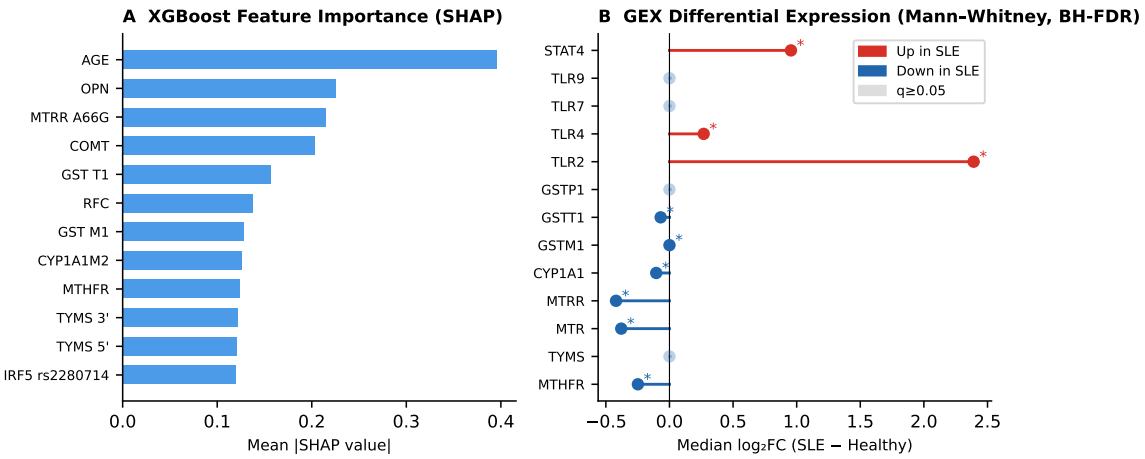


Figure 3: Feature importance and differential expression. (A) Mean |SHAP| values from XGBoost trained on SNP + covariates. (B) Median $\log_2$FC (SLE − Healthy Control); asterisks denote BH-FDR $q < 0.05$.

Discussion and Conclusion

All four pre-registered hypotheses were supported. The dominant encoding improvement (H1) is biologically expected: many risk alleles in autoimmunity act through haploinsufficiency or gain-of-function, where a single copy is sufficient to alter pathway flux. The convergent folate-pathway signal in both modalities (H2), MTRR A66G as the most significant SNP of FDR, paired with downregulation of MTR and MTRR in the transcriptome, points to reduced methionine synthase activity, decreased SAM availability, and altered DNA methylation as a mechanistic driver of SLE (Tufan et al.,). The failure of XGBoost to outperform logistic regression (H3) is consistent with the limited sample size ($n = 425$): tree ensembles typically require larger datasets to outperform regularised linear models on structured tabular data. The substantial performance advantage of gene-expression over SNP data (H4; ΔAUC ≈ 0.12) likely reflects that transcriptomics captures both genetic risk and the accumulated environmental and epigenetic dysregulation that drives active disease.

Limitations. The cohort is ethnically restricted (South Indian), limiting generalisability. The candidate-gene design of both datasets introduces ascertainment bias and variants outside these pathways are invisible. The expression dataset is multi-study and multi-platform, and although z-score normalisation partially mitigates batch effects, residual inter-study confounding cannot be excluded without explicit batch correction (e.g., ComBat). Finally, all SNP models are trained on apparent data; the small cohort size constrains statistical power and may inflate single-fold AUC estimates.

Conclusion. Pathway-level candidate-gene genotyping yields modest SLE discrimination (AUC ≈ 0.69), but the same pathway genes measured at the transcriptomic level achieve substantially higher performance (AUC ≈ 0.81), suggesting that gene-expression data better captures the downstream molecular state of active disease. The folate-cycle axis—

particularly MTRR A66G and MTR/MTRR expression emerges as the most consistently associated feature across both modalities, offering a candidate biomarker for future validation.

ACKNOWLEDGMENTS

We would like to thank Prof Meiyappan Lakshmanan for providing us the opportunity to implement our ideas through this course project

CONTRIBUTIONS

- **R Karthik:** *Team Leader* (Coordinated task allocation, scheduled meetings, Set the overall plan of action in motion, etc), involved in report preparation, identified the relevant datasets and helped with the later stages of data analysis.
- **Aldis D:** Conducted an overall literature survey, Helped in picking the problem statement, participated in the manual curation of the gene expression dataset
- **Vasudha K:** Involved in the preprocessing and normalization of data. provided the final cleaned gene expression dataset for analysis
- **Hitesh K:** Designed and implemented the entire analysis pipeline for the project

*please note that all of us contributed to the project equally in terms of hours spent on the project , ***Code availabilty*** : Project_Github

REFERENCES

Aringer, M., Costenbader, K., Daikh, D., et al. (2019). 2019 update of the acr/eular classification criteria for systemic lupus erythematosus. *Annals of the Rheumatic Diseases*, *78*(8), 999–1008.

Braun, J. A., et al. (2023). An interpretable machine learning pipeline based on transcriptomics for patient stratification in autoimmune diseases. *Patterns*, *4*(10), 100819.

Rupasree, R., et al. (2016). Application of various statistical models to explore gene-gene interactions in folate, xenobiotic, toll-like receptor and stat4 pathways that modulate susceptibility to systemic lupus erythematosus. *Molecular Diagnosis & Therapy*, *20*(1), 83–95. doi: 10.1007/s40291-015-0181-0

Tufan, A., Petri, M., Chama, L., et al. (2023). Pathogenesis of systemic lupus erythematosus: risks, mechanisms and therapeutic targets. *Annals of the Rheumatic Diseases*, *82*(8), 999–1011.